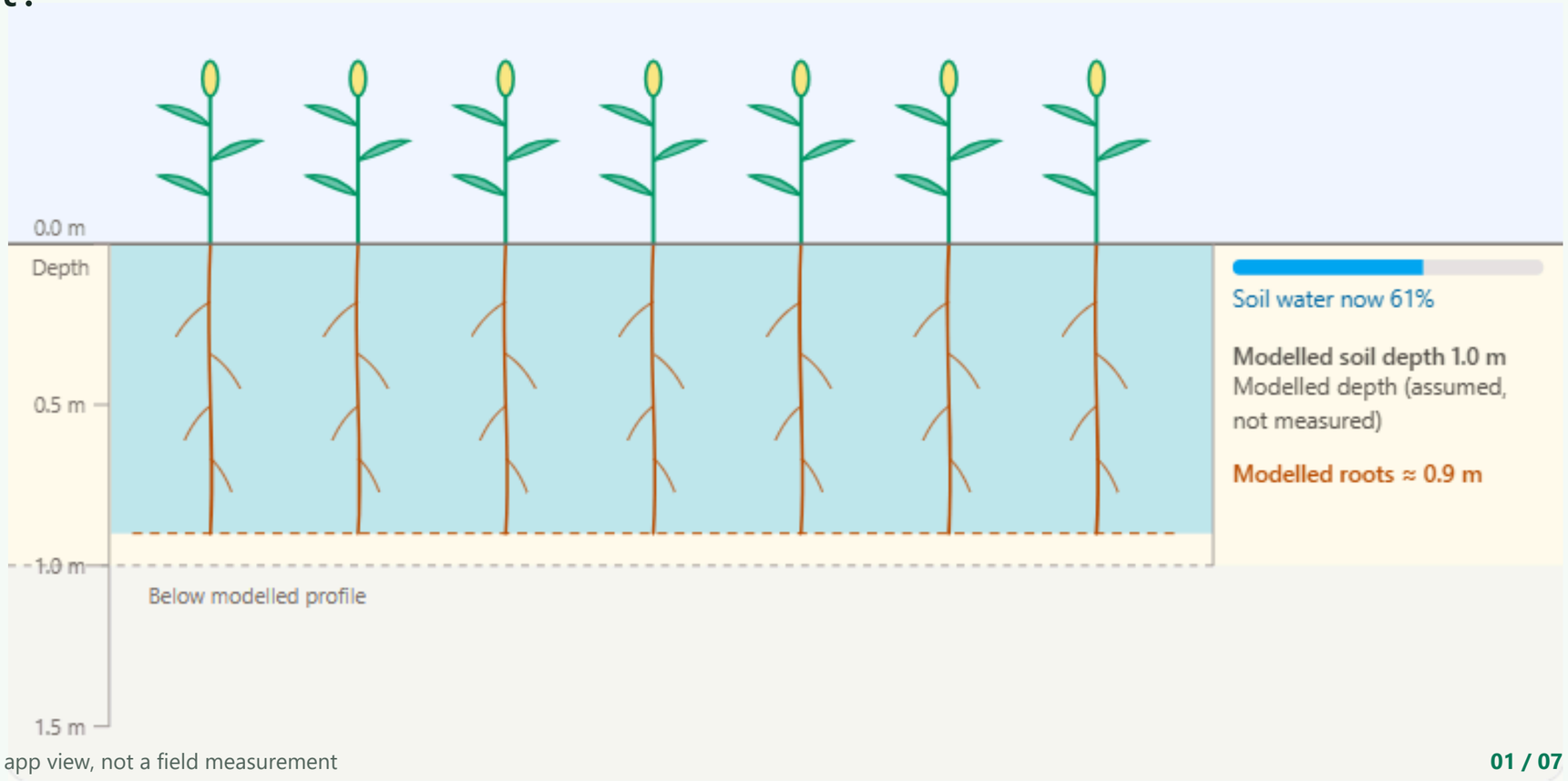
CROP-PLANNING DECISION SUPPORT

CropCompass

What should I plant next?

NASA weather. Soil context. Crop history. Your priorities.



A clearer comparison.

Explore choices across **income, water, soil and risk.**

ONE FIELD. COMPETING PRIORITIES.

A crop choice is a trade-off.

A promising crop is not necessarily the right crop for this field, this water supply, or this rotation.

01 / INCOME

Can it pay?

Consider indicative costs and outcomes in a poor season.

02 / WATER

Can it finish?

Rainfall timing and irrigation change the result.

03 / SOIL

What carries over?

Roots, residue and previous crops matter.

04 / RISK

What can go wrong?

Look beyond a rank to historical failures and assumptions.

The missing connection Turn scattered Earth data into a comparison that includes the farmer's own context.

DIFFERENT SOURCES. EXPLICIT ROLES.

From Earth data to crop options.

NASA POWER

1981+

Daily weather for historical climate and crop-water trials.

MODIS NDVI

250 m

16-day greenness history; suggests season timing, not crop identity.

SoilGrids

6 depths

Mapped soil properties; known farmer values can override them.

Farmer journal

Your field

Crop dates, residue, irrigation context and priorities.

Check + contextualize

Annual checkpoints and source labels

→ **Simulate past seasons**

FAO-style water/yield + succession rules

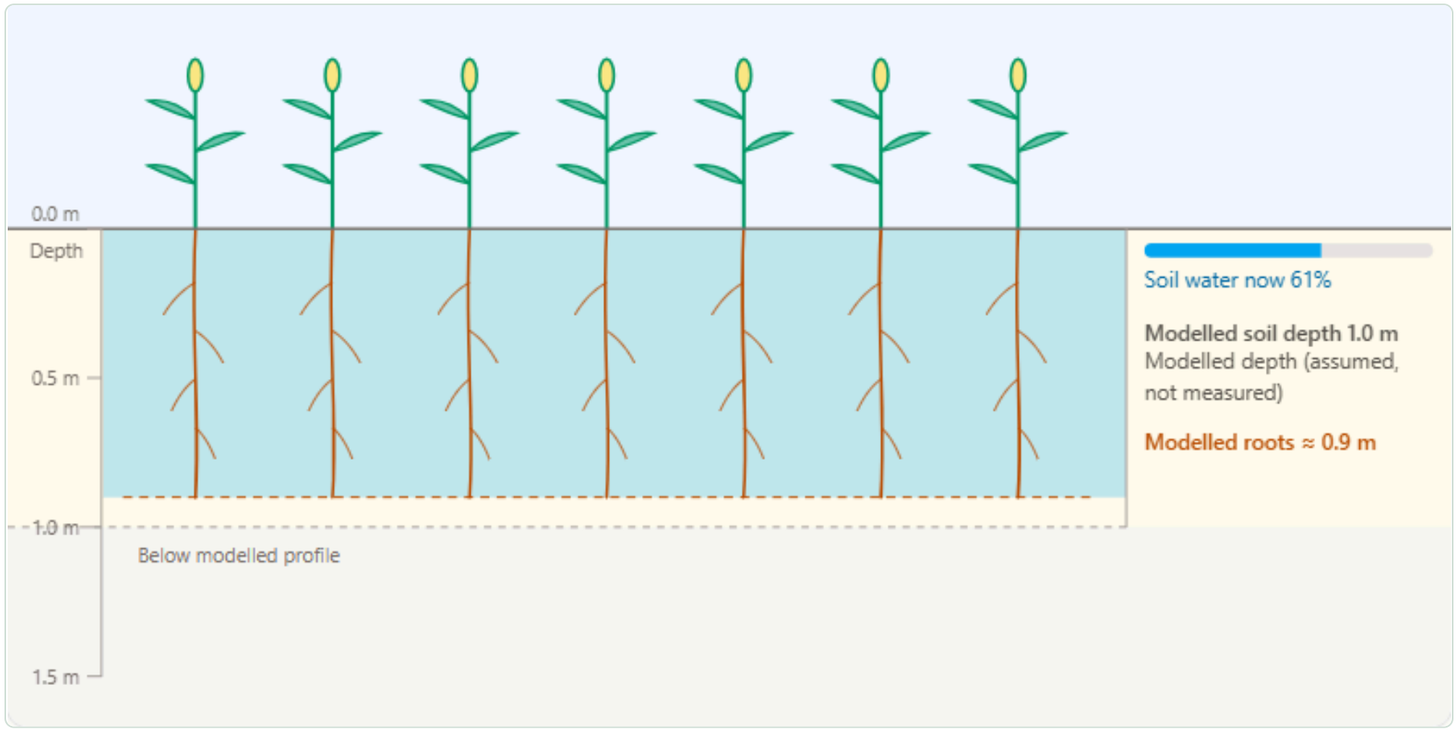
→ **Compare trade-offs**

Historical distributions, not forecasts

POWER daily/monthly APIs · MOD13Q1 via ORNL DAAC · ISRIC SoilGrids · FAO-56/33 and RothC-style methods

EXPLORE BEFORE COMMITTING

Make the assumptions visible.



Actual app capture · illustrative Telangana farm · full irrigation selected · green gram history

Try me

Preview a crop without creating a journal entry or saving a plan.

Know the source

The 1 m profile is a modelling assumption, not detected bedrock.

Keep context intact

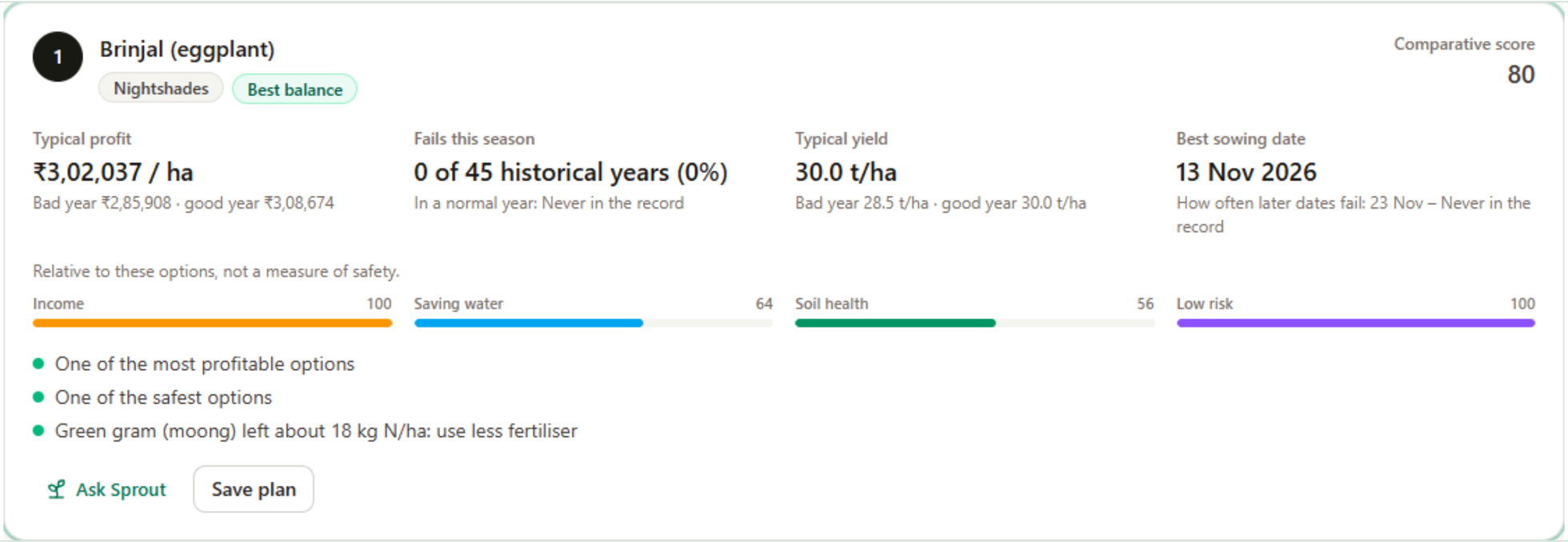
Seasonal crop fit and advice from today's modelled water are labelled separately.

The farmer can correct known soil values. The model should never hide an assumption as a measurement.

Screenshot: local app, 13 November 2026. Soil water, roots and outcomes are modelled; the sample history is illustrative.

MAKE THE TRADE-OFF EXPLICIT

Compare outcomes. Then save a plan.



Read absolute risk

Failures and yield ranges accompany the ranking.

Choose what matters

Income, water, soil and risk weights express preferences.

Think beyond one season

Explore rotations and retain assumptions in saved plans.

Illustrative full-irrigation scenario, 13 November 2026. Zero failures in the historical trials does not mean zero future risk. Prices and costs are indicative.

DESIGNED TO EXPLAIN, NOT OVERPROMISE

Useful without hiding uncertainty.

Traceable evidence

Separate source status, dates and confidence. Suspect weather stays quarantined; any historical-reference fallback is disclosed.

Verified ingestion is not field validation.

Models compute. Sprout explains.

The deterministic engine produces crop results. Sprout explains selected evidence, with consent and usage limits.

AI answers can be wrong or unavailable.



English · Hindi · Telugu

Local-language access, with terminology still needing native review.

Guest access + guided tours

Explore first; create an account to retain guest data.

Private farm records

Ownership checks, export and deletion controls.

START WITH A BETTER-INFORMED CHOICE

Better questions. Clearer trade-offs.

For farmers and advisers: connect Earth data with the field's history and the farmer's priorities.

Next: calibrate with local field observations, evaluate with farmers, and expand region packs responsibly.

crop-compass-three.vercel.app

github.com/Praannjaal/CropCompass

Data and methods

NASA POWER; MODIS MOD13Q1 V061 via ORNL DAAC REST; ISRIC SoilGrids; OpenStreetMap contributors (ODbL). FAO-56/33, Saxton & Rawls, RothC-style methods.

AI and software attribution

Development/documentation: GitHub Copilot. Runtime explanations: Groq-hosted model (source default: openai/gpt-oss-120b). React, FastAPI and other open-source libraries; full references and license inventory in SPACE_APPS_SUBMISSION.md.

Decision-support prototype. No claim of validated yield, profit, water-saving or carbon-sequestration gains. No NASA endorsement implied.